

Submission PY1-PS2-SUB01

Question PY1-PS2-Q01

scratch: show step

1. range(5) gives 0 through 4
 2. keep even i: 0, 2, 4
 3. squares => [0, 4, 16]
- => FINAL: [0, 4, 16] (check units/interpretation)

Question PY1-PS2-Q02

scratch: show step

1. counts = Counter(xs)
 2. for x in xs: if counts[x] == 1: return x
 3. return None; two linear passes $\Rightarrow O(n)$ time, $O(n)$ space
- => FINAL: scan counts, then scan in order; return first count==1 else None; $O(n)$ time, $O(n)$ space (check units/interpretation)

Question PY1-PS2-Q03

scratch: show step

1. A single pass fixes only adjacent inversions encountered once.
 2. Repeat until no swaps (or $n-1$ passes), shrinking the suffix after each pass.
- => FINAL: repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort (check units/interpretation)

Question PY1-PS2-Q04

scratch: show step

1. linear scan: up to 128 comparisons, $\Theta(n)$
 2. binary search: at most $\lceil \log_2(128) \rceil + 1$ comparisons, $\Theta(\log n)$
- => FINAL: linear $\Theta(n)$, up to 128; binary $\Theta(\log n)$, about 8 (check units/interpretation)

Question PY1-PS2-Q05

scratch: show step

1. Removing shifts later elements left while the iterator advances.
 2. Build a filtered list or iterate over a copy, then replace the original.
- => FINAL: mutation shifts elements and skips checks; use a filtered copy or iterate over a copy (check units/interpretation)

Question PY1-PS2-Q06

scratch: show step

1. range(7) gives 0 through 6
 2. keep even i: 0, 2, 4, 6
 3. squares => [0, 4, 16, 36]
- => FINAL: [0, 4, 16, 36] (check units/interpretation)

Question PY1-PS2-Q07

scratch: show step

1. counts = Counter(xs)
 2. for x in xs: if counts[x] == 1: return x
 3. return None; two linear passes $\Rightarrow O(n)$ time, $O(n)$ space
- => FINAL: scan counts, then scan in order; return first count==1 else None; $O(n)$ time, $O(n)$ space (check units/interpretation)

Question PY1-PS2-Q08

scratch: show step

1. A single pass fixes only adjacent inversions encountered once.
 2. Repeat until no swaps (or $n-1$ passes), shrinking the suffix after each pass.
- => FINAL: repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort (check units/interpretation)